Learning Social Navigation Strategies and Challenging Scenarios through Public-engaged Simulation

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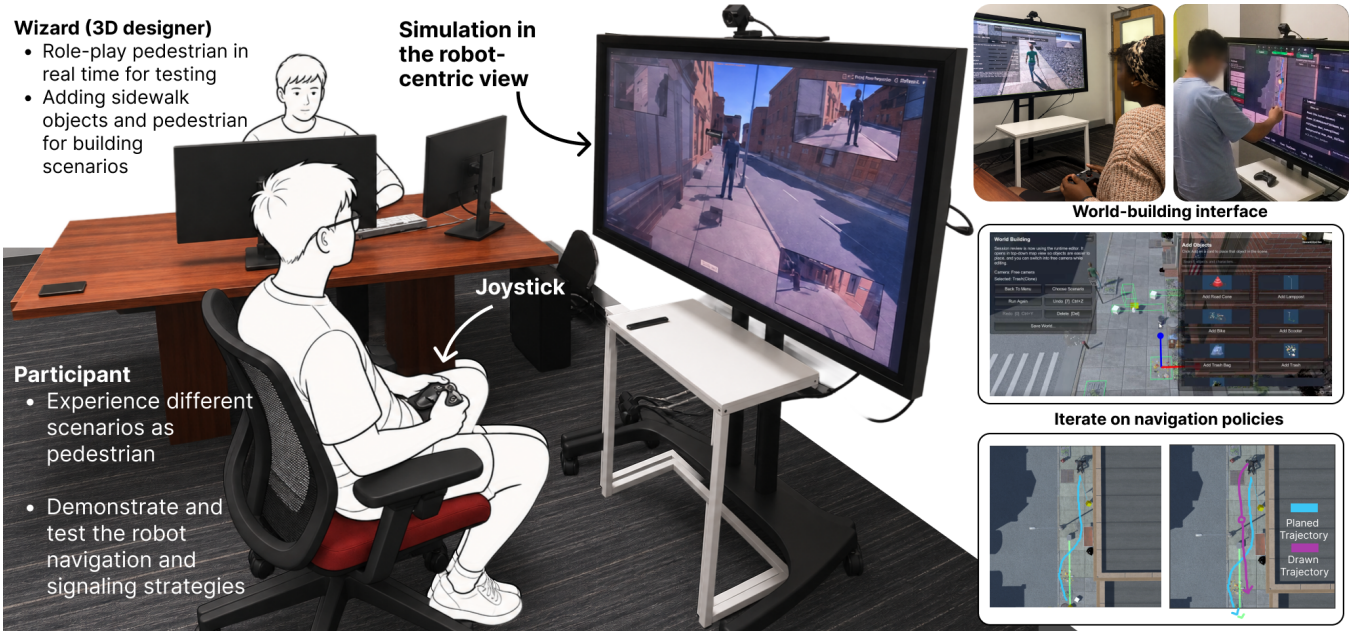


Figure 1: The participatory simulation setup. A participant drives in the simulation with a joystick, first as a pedestrian and then as the delivery robot, on a large display that shows the robot-centric view during robot trials. A researcher at the wizard station supports behavior testing and scenario editing.

Abstract

Developing public-space robot navigation requires accounting for diverse people and complex social environments. Existing simulators and datasets already seek to capture diverse human behavior in public spaces, yet people are less often engaged in situated evaluation and co-design of robot behavior. We present additions to social-navigation simulation platforms to enable effective co-design of public-space robot behavior with people. Participants experience potentially conflicting encounters with a delivery robot, then demonstrate their preferred navigation strategies by teleoperating the robot, rerunning or branching the scenario, and annotating trajectories. They can also construct sidewalk scenarios they consider important to test. In a study with 15 participants, each experienced four scenarios and contributed preferred behaviors and challenging cases, resulting in 235 navigation trajectories and 51 user-built scenarios. Their interaction traces and in-situ reflections show that the difficulty of sharing a sidewalk with a robot comes not only from its proximity but also from its unexpected behavior, the attention it demands, and its signaling. Our findings provide interactions and scenarios that roboticists developing for navigation in public space should consider.

Keywords

Human-robot Interaction, Interaction Design, Social Navigation

1 Introduction

As robots move from factories and homes into public space, they are becoming part of everyday urban life [42]. These systems serve specific stakeholders (e.g., the person receiving a delivery), but they can disrupt other sidewalk users who neither requested the service nor expect a robot [6, 42]. Training and evaluating social robot navigation therefore requires exposure to a wide range of real-world conditions, pedestrian behaviors, and situations [16].

Simulators such as SEAN 2.0 and MetaUrban [28, 52, 58] are used to train and benchmark navigation policies, usually in synthetic scenes modeled on real streets. Such simulators are often used to generate training data for navigation algorithms, in which pedestrians are modeled as crowd agents [16]. However, they are less often used to design how a robot should behave around pedestrians by engaging everyday public space users in the design process [37]. Real-world social navigation datasets such as SCAND [31] capture people's real movement around robots, though they often assume that the pedestrians found the robot's behavior acceptable. These datasets rarely contain interactions with people with different mobility behaviors in infrequent encounters that make sidewalk navigation challenging.

A different source of knowledge can help fill this gap—how people talk and think about their experiences on sidewalks. Such experiences include how people reason about what kind of scenarios are difficult to manage on a sidewalk and what they would consider acceptable and socially compliant robot behavior [22]. Participatory design in human-robot interaction (HRI) has long been used to draw out this kind of tacit knowledge and interaction expectations from end users [5]. Prior work has paired participatory design with simulation to test alternative robot behaviors and co-construct preferred trajectories [8, 50], showing that participants can surface breakdowns and propose recovery behaviors that designers had not anticipated. These tools, however, have mostly been applied to robotics cases involving facial expressions and body motion, and have not been connected to robots operating in public space.

In this work, we revise SEAN 2.0, an open-source simulation platform originally built for benchmarking robot navigation policies [52], to be used for engaging people in experiencing and redesigning socially compliant robot navigation in public. We focus on sidewalks because they are a daily yet challenging setting where end users already report conflicts with robots, particularly around accessibility [21]. We extend SEAN 2.0 with end-user-facing interaction mechanisms, world-editing tools, and communication prototyping features so that end users can experience, evaluate, demonstrate, and co-design robot navigation themselves. With this participatory simulation tool, we ask two research questions:

RQ1 What social navigation strategies and signaling preferences can we learn from end users who experience sidewalk encounters in simulation?

RQ2 What interaction challenges and scenarios do end users construct when asked to make sidewalk navigation harder?

To answer these questions, fifteen participants each experienced four sidewalk scenarios twice, first as a pedestrian and then as the delivery robot. They replayed and annotated their own trials and rebuilt each scene to make it more challenging and thus more likely to cause issues between robots and people (Sec. 4). Through iterative testing and scenario building, participants' evaluations surfaced the challenges a social navigation policy may face. Participants also proposed navigation and signaling strategies that could be more socially appropriate, along with the reasoning behind them (Sec. 5). These findings ground future directions for participatory simulation and for learning socially compliant navigation (Sec. 6). We contribute (1) a social navigation simulator extended for participatory co-design of robot behavior, and (2) evidence that design choices participants make independently but in common, such as scenarios built at the same spots, indicate where people expect public-space robots to fail.

2 RELATED WORK

2.1 Social Navigation Simulators and Datasets

Social navigation research has produced many platforms for modeling crowds and evaluating socially aware navigation [15, 37]. Beyond general-purpose simulators such as Gazebo and Isaac Sim, dedicated platforms target social settings. SEAN 2.0 formalizes and generates social situations [52], SocialGym and Arena 4.0 provide benchmarking pipelines [28, 44], and MetaUrban builds large urban scenes with parallel training [58]. Most simulated pedestrians follow variants of the social force model [26], and evaluation guidelines emphasize metrics on goal-reaching success rate and collision avoidance [16, 17, 53].

Alongside simulators, several datasets supply human demonstrations for learning social navigation. SCAND records expert teleoperation of two robots through a campus and downtown streets [31], and ACME records roughly 30 hours of data in socially rich environments such as dining halls [35]. Other efforts collect demonstrations in virtual reality [12] or record the forces between pedestrians and robots to fit better pedestrian models [3]. Recent robot learning methods also use positive and negative demonstrations to learn preferences [23, 33, 56]. However, these datasets mostly capture what an operator controlling the robot chose to do, not why they chose to do so or any alternatives that could be more socially compliant. They also have limited challenging encounters, such as a blocked curb cut or a head-on meeting on a narrow sidewalk. Both of these situations, among many more, are still open challenges for social navigation [16, 37].

2.2 Co-Design for Robot Behaviors

Participatory design in HRI involves stakeholders in shaping robot behaviors from the earliest stages of development [5, 41]. Methods range from design canvases that structure early ideation [5] to participatory prototyping that aligns robot behaviors with how users perceive them [10] and situated sessions with robots operating in the intended domain [48]. For public-space robots, co-design has been used to elicit user expectations [51], confront stakeholders with sidewalk conflicts involving delivery robots [55], and gather design insights from people with mobility disabilities and robotics practitioners [9, 22].

Simulation has been used to prototype experiences that may be difficult to capture in real-world settings. Web-based and virtual reality environments make remote robot design studies engaging and accessible [8, 43, 50]. Related traditions in Human-Computer Interaction and automotive research demonstrate how simulated humans and environments can stand in for costly field deployments [20, 49], and how replaying recorded interactions from multiple perspectives can support reflection on what happened [19]. However, these participatory simulation tools have rarely been connected to the simulators used in robot learning to develop and evaluate navigation policies. This leaves an open question of how participatory simulation can be used to produce interaction data and scenarios that can inform robot learning and evaluation.

2.3 Sidewalk Robot Navigation and Signaling

Field studies of sidewalk delivery robots document how pedestrians notice, avoid, and sometimes assist them [1, 13, 54]. Observations from pilot deployments and surveys of pedestrians and bicyclists show that most encounters are uneventful, but people's comfort levels and potential safety issues vary with how the robot moves near them and where it stops [18, 57]. These challenging situations, though uncommon, can lead to issues that inhibit people's movement on already crowded sidewalks [6]. These encounters are shaped by the social and material order of public space, not only by the robot's trajectory [40, 42]. Problems can be most acute for people with visual impairments or motor disabilities. Prior work

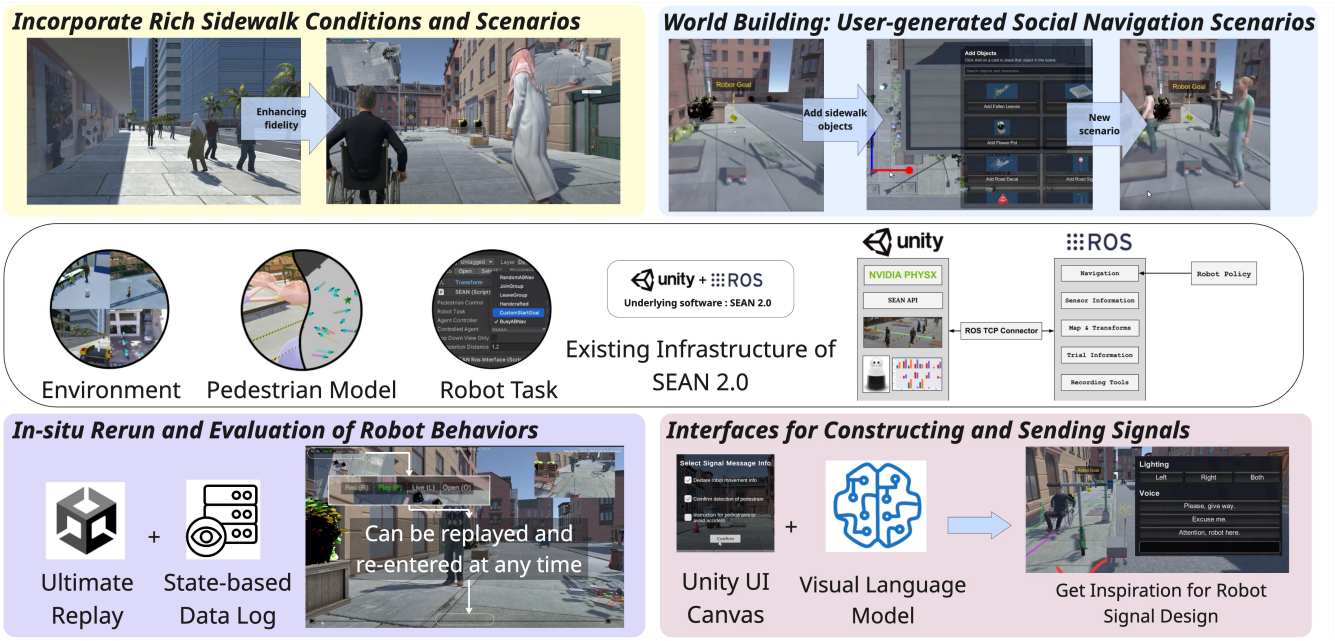


Figure 2: Extensions to SEAN 2.0 for participatory simulation. Around the existing Unity–ROS infrastructure (center), four additions support richer sidewalk environments and pedestrian avatars (top left), participant-guided world building (top right), state-based replay and branching reruns (bottom left), and interfaces for prototyping robot signals with suggestions from a vision-language model (bottom right).

with these populations reports how robots can block people's paths or behave unpredictably around them [7, 21].

To avoid issues, robots can signal intent during challenging encounters [59]. Prior work has studied light and screen indicators [14, 45], voice communications [25], projected paths [11, 46], external displays [30], and nonverbal motion cues [4, 24, 27]. A scoping review shows that such cues can increase legibility and reduce ambiguity [38]. While signaling can be an effective technique for improving social navigation around people, no existing simulator allows users to design and test communication together with navigation. Thus, it is challenging to design composite navigation and signaling strategies for improving social navigation. Our work specifically aims to allow people to co-design both strategies as a way to inform future robot design.

3 SYSTEM DESIGN AND IMPLEMENTATION

To adapt a navigation simulator for broad community participation, we modified SEAN 2.0, an open-source Unity—ROS platform for evaluating social navigation in crowd environments that already supports autonomous navigation [52]. What SEAN 2.0 lacks for participatory use is scenario control and a usable interface for end users to interact with the robot. Figure 2 shows our four additions alongside the existing infrastructure. A demo video can be found in the supplementary material. We plan to release the simulator source code and study data upon acceptance.

Sidewalk fidelity. The original simulator pairs a social-force pedestrian model with a flat, generic sidewalk. We integrated a high-fidelity streetscape from the Unity Asset Store[1] with curb cuts, pavement irregularities, and street furniture, and diversified the pedestrian population with wheelchair users, dog walkers, and scooter riders by empirically re-tuning the social-force parameters for each avatar [26].

In-situ rerun. Social navigation involves fleeting moments that participants may want to revisit and re-enact. We added timestamped state logging that allows branching reruns [19] from any selected moment. Users can pause where robot behavior felt wrong, examine how the autonomous policy unfolded, then retry alternatives or teleoperate the robot to demonstrate the behavior they prefer. This is further supported by a drawing mechanism through which they can directly draw trajectories and run them to compare with automated algorithms.

World building. Navigation edge cases are rare and hard to anticipate, and SEAN 2.0's fixed tasks do not give users control over the scene. We developed an interface for users to pause the simulation and rearrange furniture, reposition pedestrian agents, or construct new sidewalk layouts that reflect their lived experiences. Built scenarios are saved locally for reuse and comparison. We built an initial sidewalk object library and can also generate new objects on the go using Meshy.AI, a text-to-3D platform [2].

Robot signaling prototyping. A signaling interface (Fig. 5) lets participants pause navigation and state what the robot should

[1]Resource: https://assetstore.unity.com/packages/3d/environments/urban/street-new-york-183319
[2]https://www.meshy.ai/zh

communicate using voice or light. They can also draw inspiration from a vision-language model. The interface pairs the stated goal with a robot-centric RGB image and passes both to Gemini 3.5 Flash, which returns candidate text and audio signals that can be refined and replayed in the scenario through the rerun mechanism, helping participants put their communication preferences into words.

Setup and device. Sessions ran in a lab environment (Fig. 1). We used a large display so that the robot could be more easily viewed. The participant drove as a pedestrian avatar and then later as a robot with a game controller. The controller supports both analog sticks and hotkey controls. During robot trials, the display showed the robot-centric view, with first-person, third-person, and rear cameras available. Behind the participant, a researcher at a two-monitor wizard station monitored the scenario and placed objects during world building.

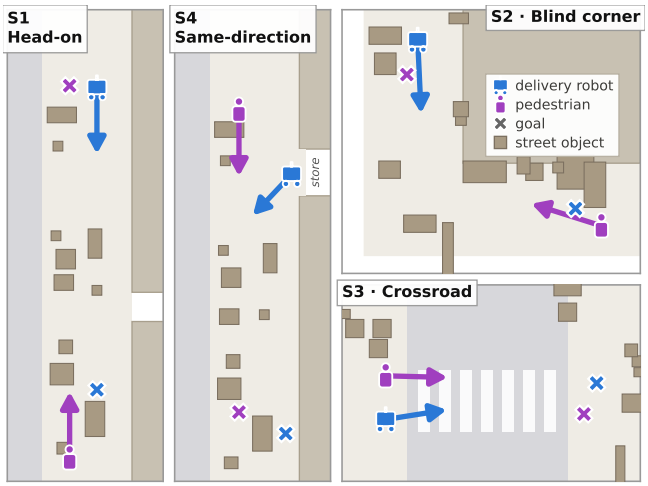


Figure 3: The four sidewalk scenarios (schematic, approximately to scale). Icons mark where the robot (blue) and the user-controlled pedestrian (purple) start, short arrows show their initial headings, and × marks their goals.

4 User Study

Using the revised simulator, we conducted a study with 15 participants recruited from our university community. Twelve participants were students, and three were university staff. Their mean age was 22.82 years (SD = 4.02). P15 self-identified as a wheelchair user, while the other participants did not report any mobility disabilities. Their full demographic information can be found in the supplementary materials. Participants were compensated $30, and the study was approved by our institution's IRB.

Each participant experienced the four sidewalk scenarios we pre-designed in a random order: a *head-on encounter* on a narrow sidewalk, a *blind-corner encounter*, a *crossing encounter*, and a *same-direction encounter* (Fig. 3).

Procedure. Each session lasted 75 minutes and began with a practice scene in which participants familiarized themselves with the control joystick and viewpoints by navigating to goal markers as both the pedestrian and the robot. For each scenario, participants first completed the trial as a pedestrian and then as the robot. After each trial, we replayed the recorded trajectories and conducted an interview asking participants about comfort, path adjustments, navigation rationale, and preferred alternatives.

Participants could also sketch an alternative robot trajectory directly on the display, which the simulator then executed as a further trial. Participants then co-constructed new situations (world building) by adding or rearranging objects and characters and by adjusting the scene's existing agents (speed, yielding behavior, and erratic behavior). We then ran these modified situations as additional trials. Sessions concluded with overall reflection questions.

Measures. We also collected think-aloud interview data in all fifteen sessions. After each pedestrian trial, participants rated two items on 1–10 scales: how comfortable they felt sharing the sidewalk with the robot (1 = uncomfortable, 10 = comfortable) and how much they had to adjust their path to accommodate it (1 = not at all, 10 = completely). These ratings were primarily exploratory and helped us understand the level of disruption participants experienced. Positions were logged at 10 Hz for all agents. This yielded 235 robot runs (87 under the autonomous planner, 80 teleoperated, 38 following a drawn trajectory, and 30 re-runs of edited worlds).

Analysis. The first two authors inductively coded participants' think-aloud comments, resolving disagreements through discussion and iteratively refining and grouping the codes into themes. Ratings and trajectory measures were summarized descriptively.

5 Findings

5.1 How do participants perceive sharing a sidewalk with robots in the simulation?

We observed substantial variance in the two measures after participants navigated as pedestrians. Comfort ratings varied most in the blind-corner scenario (σ^2=7.81), while path-adjustment ratings varied most in the head-on encounter (σ^2=7.21). The corner was no challenge at all for some (P7 and P10 rated comfort 9 and 10) and the most disturbing moment of the study for others (P6 rated 2). From the logged trajectories, we saw that in the head-on scenario, where the participant and the robot walked straight toward each other, a typical participant slowed to about half of their normal walking speed when the robot was closest (0.48× at the median) and came to a full stop about four times in a single trial.

After each run, we replayed the trajectories and asked the participants about their navigation experience and perceived disruption. The most prevalent disruption was being pushed to an area of the sidewalk they would never normally walk on. P6's avatar stepped onto a hatchway on purpose to escape the robot, scoring comfort 2 with adjustment 10. P9 named the trade-off between passing speed and comfort, *"If I need to get to my destination now, I'm more likely to step on something I wouldn't normally step on."* Asked whether giving way like that would frustrate them, they answered *"Probably. Only because it's not a human."*

Some participants also did not adjust their path because they assumed the robot would yield. P12 reasoned that *"this kind of robot has a camera"* and *"[I] just assumed that the robot can detect me and avoid hitting me,"* so they proceeded, though they admitted

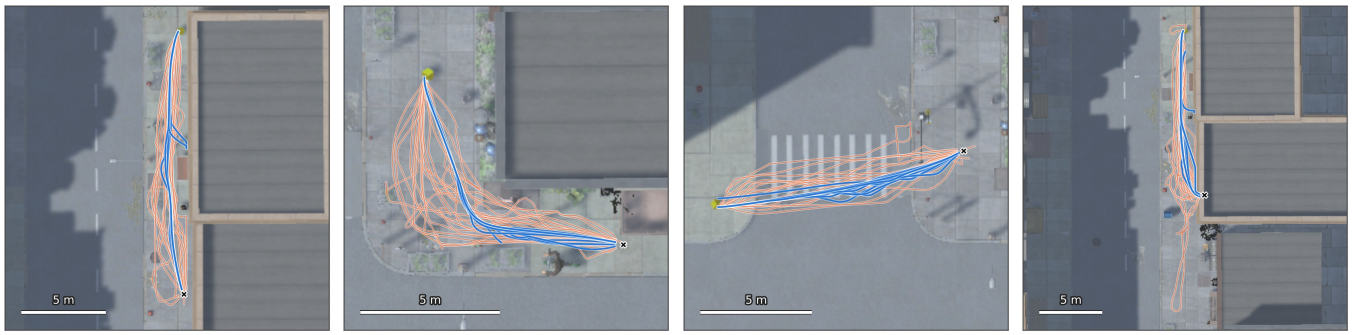


Figure 4: Autonomous policy versus user-teleoperated trajectories. Blue shows the autonomous planner, while orange shows the participant-driven runs. The planner repeats essentially one trajectory per scene, while participant-driven trajectories fan out around it, concentrating their disagreement at social encounters.

that they kept watching it. Comfort and disruption also diverged in participant ratings. P5, for example, adjusted their path *not at all* at the crossroads yet rated comfort 4. They never had to move, but they watched the robot the whole way, *"trying to figure out which way it's gonna go, so I don't actually run into it… the robot starts facing me, and we're sort of close to each other."* In this sense, keeping the robot in view was itself a cost. The simulation interaction traces and in-situ review mechanism helped reveal less visible sources of discomfort that were not apparent from the trajectory alone.

Using this simulator, we repeatedly observed participants yielding early and found that they did so because they did not trust the robot to react appropriately. P4 stated, *"I wanted to avoid the robot… as opposed to having it recognize me and move,"* because *"when walking towards a robot, it's a little bit more unclear how the robot would behave."* Much of the disturbance came not from the physical effort of yielding, which participants routinely do for other people, but from not knowing what the robot would do (P8, *"I can't predict where it's actually gonna go"*).

Participants also built different mental models of the robot: as a vehicle (P1's *"mini car"*), a pet (P9's *"dog off the leash"*), or another pedestrian (P7, P12). Their framings of the robot influenced which rules they applied when evaluating the robot's behavior. P7, holding the pedestrian frame, rated every scenario no worse than a human doing the same thing, while those who saw the robot as a vehicle applied traffic rules when evaluating its behavior.

5.2 What social navigation strategies and considerations were revealed?

5.2.1 Balancing navigation efficiency and path legibility in public space. To quantify similarity between trajectories, we computed the discrete Fréchet distance and the symmetric mean nearest-point deviation for every pair of runs within the same scene, using the raw 10 Hz position logs. Participants diverged more from one another than from the planner, with participant-to-participant distances (median Fréchet 1.6 m, mean deviation 0.55 m) exceeding participant-to-planner distances (1.13 m and 0.39 m). This suggests that participants interpreted social compliance differently and that there may not be a single unified trajectory.

One difference in reasoning underlying these trajectory choices was how participants balanced efficiency against the legibility of the robot's motion. In the blind-corner scenario, four participants (P6, P8, P10, P11) chose a slower line that kept the robot visible rather than taking the shortest path. P8 and P10 drove the wide arc from the start; P11 split the turn into two straight legs with a pivot in place so that each leg was easy to anticipate. P5, who took the inside line, recanted (*"socially it would have been more acceptable to swing out wider"*). P8 stated that the robot's path should have *"larger curvature so the human can see me."*

Table 1: Autonomous planner vs. participant driving

	Stopped ≥1 s near human	Stopped (s)	Speed near/free	Path eff.
Auto policy (DWA)	0.00	0.4	0.91	0.99
User demonstration	0.62	4.1	0.83	0.94

Speed is the mean speed near a human (within 1.5 m) divided by the mean free speed, and efficiency is the median. Trailing idle time at the goal was excluded from the stop measures.

5.2.2 Stopping smartly as a sign of socially compliant navigation. Table 1 compares the autonomous planner with participant driving. SEAN's default planner (DWA) never stopped for as long as a second near a human in any autonomous run. ***Participants, in contrast, stopped routinely and treated stopping as a preferred behavior for social navigation.*** They averaged 0.62 stops of at least one second near a human per run and stood still for 4.1 s per run, compared with 0.4 s for the planner. While away from people, their trajectories followed the planner's closely (with 68% of the driven trajectory within half a meter of the planner's). User demonstrations and the autonomous policy differed mainly where the robot is close to a pedestrian (Fig. 4), where the planner mostly kept going. Participants described these stops as giving the person the right of way. While driving the robot, P8 assumed that pedestrians *"were not gonna give way to the robot, so I was just gonna wait,"* and P4 called it the default while navigating around people, *"it's better to just stop and let them pass by."* Where participants stopped depended on predicting the pedestrian's path and staying out of it. P3 stopped the robot *"to the side, so I could see what they're gonna do"* and P10 chose *"not the quickest, but the angle that creates the least disruption."*

Participants also put a cap on waiting time. P13 said *"in the real world, there'll probably be more pedestrians, so it would probably be unrealistic that [the robot] just keep waiting."* P8 echoed what the robot should do once the waiting time runs out, stating that it should *"wait a couple seconds, signal; if nothing happens, take the other route."* The simulation helped participants visualize stopping strategies and further reason about when waiting should end and what the robot should do next.

5.2.3 Reasoning around social norms given different sidewalk conditions. The simulator also let us watch how participants reasoned about social norms as conditions changed. In the crossroad scenario, the countdown of the crossing signal and the turning cars made participants use the pedestrian as a guide. A walker ahead of the robot cleared the path and showed drivers that someone was crossing, so P8 chose to *"trail it, copy what it does,"* and P11 applied the same rule (*"if the person goes, I follow them"*).

Participants revised their strategies as they tested and replayed. P3 changed their yielding policy as soon as they realized the storm drain grate could trap their wheels (*"I could theoretically get stuck in that"*). With one pedestrian present, P5 cut diagonally across the sidewalk, but reasoned that *"if there were more people, it would probably be better to maintain a constant speed and stay on one side of the sidewalk."* After adding more pedestrian avatars to the scene, they further recognized that navigating a crowd required them to wait for openings rather than simply maintain a speed.

5.2.4 Accessibility concerns and driving around mobility-aid users. Participants also reasoned about who else uses the sidewalk and could be disrupted by the robots. P13 said they would drive the robot differently *"depending on whether [the pedestrian was] on wheels or not."* P9, while driving the robot, reflected that *"even a person who needs to use a cane… they need the flat surface, because that's going to give them the best chance where they're not going to fall."* The flat part of the sidewalk is what both the robot and a person with a mobility aid need, and participants who noticed this gave it to the person. P10 rebuilt a case they had read about (likely [2]), in which a robot parked on a curb cut while a wheelchair user's crossing signal ran out. P15, who uses a wheelchair, validated the tension around the flat surface, which could be important to both them and the robot. They also added another challenging scenario, stating that a steep curb cut cannot be taken head-on in a wheelchair, so a robot that forces its approach angle can be in their way without touching them. P15 wanted the robot to hug the wall in the head-on scene so that a person in a hurry could take the storm drain grate.

5.3 How do participants want the robot to communicate?

The need for signaling often arose when participants felt that the robot needed to communicate intentions to nearby pedestrians that were difficult to convey through motion alone. For example, six participants (P4, P6, P8, P10, P12, P13) wanted the robot to speak or beep when a pedestrian blocked its path and waiting did not resolve the situation, four (P3, P5, P8, P9) asked for a turn indicator like a car's blinker so people could tell where the robot was heading, and two (P10, P12) wanted a stopped robot to say *"I'm waiting for you"* so that a yield could be distinguished from a breakdown. They did not converge on a single signal type, but they did suggest when to signal. Visual signals can be useful when the robot is in their field of view, sound can be useful when it comes from behind (P8), and light can be useful in loud environments (P12).



Figure 5: The signal-sending interface. One scenario in which participants thought the voice signal was useful was when a person blocked the path to the robot's destination.

Participants also came up with a range of designs for the signals themselves, and these designs sometimes conflicted with one another. *"Excuse me"* is a rude presumption to P2 and P4, yet perfectly welcome to P12 and P13. Voice is *"creepy"* to P2 and *"weird"* to P10, yet delightful to P12, who exclaimed, *"oh! The robot is talking!"* P4 offered the synthesis of *"a robot voice, but a kind robot voice."* Preferences also differed on whether the robot should travel in front of or behind people. P4 and P11 wanted the robot to travel behind (*"I can just forget about it"*), while P6, P8, and P9 wanted it visible in front (*"I don't trust it enough when I can't see it"*). Participants also pointed out what a signal cannot fix. A signal only works if people comply with it (P8), if they can tell which robot sent it when several robots share a block (P10), and if they can perceive it at all, since deaf pedestrians need the visual channel and blind pedestrians the audio one (P14). P8 also warned against overusing signals: *"imagine if the robot were to hog that path and start blaring 'make way, make way.' That's kind of annoying."*

5.4 What are some challenging scenarios that they came up with?

When asked to make our pre-defined scenarios harder, participants placed 212 objects across 51 edited worlds (Fig. 6). Of the objects placed, 57% were pedestrian avatars with specific identities (pedestrians, cyclists, wheelchair users). Twelve of the fifteen participants placed mobility-aid users unprompted (27 wheelchair, mobility-scooter, cane, and walker users), a reminder that sidewalk robots share the walkway with people who move differently from the able-bodied pedestrian most navigation planners assume.

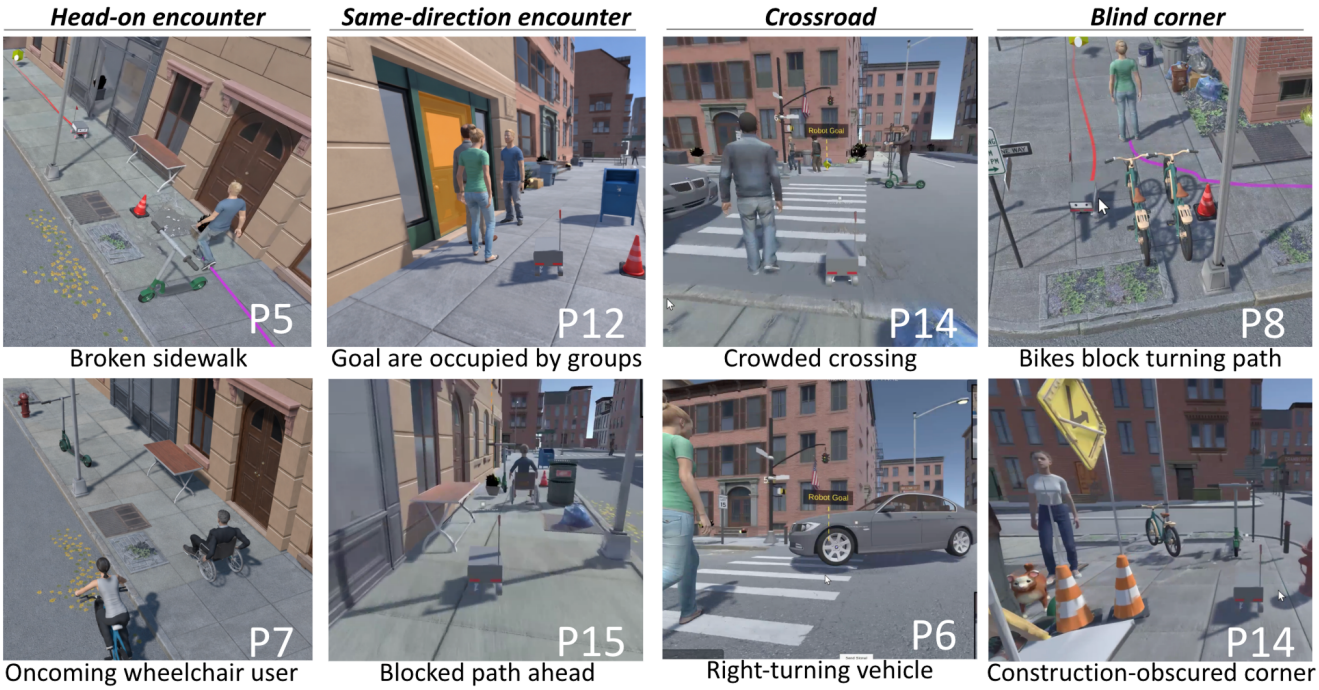


Figure 6: Eight examples of participant-built scenarios, arranged in two rows. Each column shows two user-generated variations of one of the four original scenarios.

The sidewalk object placements converged spatially, even though no participant saw another's session. In the blind-corner scene, 11 participants wanted to place objects at the corner. The corner edits shared the intent of blocking the sightline to the other sidewalk so that the robot and pedestrian would meet with no advance warning.

Another common pattern was narrowing or blocking the path where the robot met people. In the head-on encounter, P2 narrated the design as they built it, *"put something right opposite the table, where there's only space for one person, and we get there at the same time."* At the crossroads, participants added a car turning into the crossing, and the researcher drove the car in real time to produce it (see the right-turning example in Fig. 6, bottom row, third column). Driving the robot through that edit, P8 identified the problem as soon as the car appeared, *"I wasn't sure if the car would have seen me as the robot, so I was just gonna wait until it completely passed. I don't know if the car's gonna see the small robot."* Participants also staged specific social dilemmas. P11 reconstructed scaffolding whose pole rows *"the robot can pass under but a person can't."* P10 recreated a reported case of a robot camped on a curb cut while a wheelchair user's crossing timer ran out. P9 placed cones because they get kicked and fall into the walking path, and P6 asked for the study's only weather edit, rain, *"because it can be hard to see the robot."*

6 Discussion

6.1 Social Navigation Simulation as a Participatory Design Medium

Participatory design for public robots has so far relied mostly on workshops and storyboards [9, 22, 51, 55]. Our study shows that, with thoughtful revisions, a simulator typically used for benchmarking can serve as a design medium. Participants' iterative testing loops can help us answer why questions and prototype what-if scenarios within a single session, without building a new study around each one. Replay [19] let participants reconsider a choice they had just made, prompting reflection on the scenario. World building turned lived experiences and public concerns into test cases. In our setup, the robot also runs an autonomous policy, which naturally serves as a baseline for seeing how users control the robot differently and what kinds of socially intelligent behavior it needs to exhibit.

Looking ahead, we see participatory simulation supporting more prototyping of robot behavior in social navigation, in the same way that driving simulators have supported rapid iteration on vehicle behavior with the people it affects [19]. It can also serve as a data collection tool. For example, the simulator can load different baseline policies and let end users experience each one and evaluate the differences, so that the positive and negative examples they produce are labeled against a known baseline. There is also room for tighter integration with real-robot experiments. For example, a scenario built and rated in simulation can be replayed on a physical robot in the real world. The gap between the two ratings can be a measure of how much the simulation can be trusted [29]. In social robot design more broadly, where the same encounter looks different to the robot and to the person it passes [37], an open challenge for platforms like ours is keeping those two views tied to the same moment so that the operator's demonstration, the pedestrian's rating, and the replay all refer to one event.

6.2 Enhancing Scenarios and Pedestrian Behaviors in Simulation

The scenarios participants built are a direct input for benchmark simulators. Platforms such as Arena and MetaUrban evaluate policies on generated or replayed scenes [28, 44, 58], while covering rare encounters is a general guiding principle [16]. Our 51 edited worlds uncovered challenging scenarios such as blocked sightlines at corners, single-width passages, and cars turning into a crossing. Those scenarios, especially the ones that different participants were all concerned about (e.g., navigating around people using different mobility aids), can be viewed as indicators of where they believe public-space robots may fall short and thus may warrant more consideration before deployment.

The pedestrian model is another major gap. Fourteen participants reasoned about wheelchair users, cane users, and cyclists. Even when the avatars look plausible, studies with people with motor disabilities show that the hard cases depend on details such as turning radius, the room a chair needs, and whether a curb cut is the only way across [21, 22], which current crowd models struggle to capture [26]. The next step could be to run our protocol with community members whose mobility needs are missing from current datasets so that they can demonstrate how they navigate around robots. This echoes calls to design public robots *with* the people who share the sidewalk rather than *for* them [34, 36].

Finally, the shared design choices among participants could be high-value interactions that robot developers should pay more attention to [49]. For example, field studies report that comfort varies with how the robot moves and where it stops [18, 54, 57], which matches our simulation results. However, we view our simulation as a tool for surfacing interaction challenges and testing alternative robot behaviors. The resulting findings remain hypotheses requiring field validation, since simulation lacks the robot's physical presence and the real consequences of its actions.

6.3 Learning Socially Compliant Navigation and Signaling

Existing demonstration datasets record what an operator did while driving a robot [12, 31, 35]. Our work additionally captures why members of the public acting as operators rejected particular behaviors and what they preferred instead, because every rerun pairs an abandoned behavior with a preferred one from the same moment. The accompanying reasoning and multifaceted examples are what recent robot learning methods call for, including learning from positive and negative demonstrations [23, 33] and inverse reinforcement learning of implicit social behavior [32].

The findings also speak to what should count as socially compliant. Participants' trajectories were less similar to one another than they were to the planner's. This suggests that robots should have a set of readable trajectories, not one expert trajectory. A learned policy should therefore represent a distribution of trajectories from which it can choose based on the pedestrian in front of it, making adaptivity, rather than imitation accuracy, a central objective. Most social navigation work still scores policies based on speed [16], while our results show that stopping strategically is what participants considered socially compliant. The recurring rules participants stated (follow the pedestrian ahead and wait with a cap before signaling and rerouting) also call for further development and testing of the robot's social intelligence.

For signaling, prior studies tested single cues, such as lights, projections, or displays, one at a time [30, 38, 45, 46]. Our participants instead gave a rule for choosing among them: match the channel to what the pedestrian can perceive. This rule can be tested directly in the simulator by exposing several signal channels at once and restricting what the pedestrian avatar can sense. Vision-language navigation methods that reason about social context from the robot's view [39, 47] are a natural next step toward such joint motion-and-signal policies, since the same model that judges the scene can also pick the signal.

7 LIMITATIONS AND FUTURE WORK

Our study has four main limitations. First, we recruited 15 participants, mostly students, and only one used a mobility aid. Future work should build on this protocol for broader community engagement with older adults, people with disabilities, and residents of neighborhoods where sidewalk robots already operate. Second, although we ran the pedestrian view on a large screen, sitting in front of a screen is a different embodiment from walking, and this may have shifted what felt too close or too late. A head-mounted display or a walk-in setup would bring the study closer to the street. Third, the simulation removes the robot's sound, its physical presence, and the cost of a collision. Future work would benefit from testing the insights we collected here with a physical robot on real sidewalks. Finally, we found that some participants were using a joystick and a first-person camera for the first time, so we provided a practice run before the scenarios and let users tune the joystick sensitivity themselves. Future participatory simulation platforms need accessible input methods and appropriate pacing when conducting studies with mobility-aid users.

8 CONCLUSION

This work illustrates how extending a social navigation simulator with world editing, rerun branching, and signaling prototyping can turn it into a participatory design tool. Across fifteen sessions, the simulator let us see the disruption a robot introduces from the pedestrian's perspective. Through participatory sessions, we learned the mental cost of an unpredictable robot and how far it may push people off their everyday walking path. We also learned how participants reasoned about the robot's navigation and how its navigation can become more socially intelligent. For example, the robot can trade efficiency for legibility at blind corners, stop to give pedestrians the right of way, and choose where to stop by predicting the person's path. The harder scenarios participants built also demonstrated unmet needs in current social navigation datasets. Together, these findings show that a simulator built for policy evaluation can also be a place where the public tells us what a socially navigating robot should do, and where those answers become demonstrations and scenarios that robots can learn from.

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